Goodness of fit test

Categorical Data

Definition: When each observation can be classified into one of finite types or categories.

Examples:

- types of credit cards accepted in a store (Visa, MasterCard, AmericanExpress)
- blood group of students in a college (O, A, B, AB)
- degree of agreement with a statement on a questionnaire (strongly agree, agree, neutral, disagree, strongly disagree, etc.)
- grade in an exam (such as A, B, C, etc.)
- patient condition (poor, fair, good, excellent)

Test of hypothesis involving categorical data : Example

We want to test a hypothesized proportion of people in a population belonging to blood groups A, B, AB and O based on random sample drawn from the population.

Table 10.1 Counts of blood types for white Californians

A	B	AB	O
2162	738	228	2876

Table 10.2 Theoretical probabilities of blood types for white Californians

A	B	AB	O
1/3	1/8	1/24	1/2

Test of hypothesis involving categorical data : Multinomial Experiment and Hypothesis to test

A **multinomial experiment** satisfies the following conditions:

1. The experiment consists of a sequence of n trials, where n is fixed in advance of the experiment.
2. Each trial can result in one of the same k possible outcomes (also called categories).
3. The trials are independent, so that the outcome on any particular trial does not influence the outcome on any other trial.
4. The probability that a trial results in category i is p_i, which remains constant from trial to trial.

The parameters $p_1, \ldots, p_k$ must of course satisfy $p_i \geq 0$ and $\sum p_i = 1$.

$$H_0: \quad p_i = p_i^0 \quad \text{for } i = 1, \ldots, k, \qquad p_i^0 > 0$$

$$H_1: \quad p_i \neq p_i^0 \quad \text{for at least one value of } i.$$

N_i: number of samples in category i

N_i>0

$\sum_{i=1}^{k} N_i = n$ where n is the sample size

Obs:

$(N_1, N_2, \ldots, N_k)$ follows multinomial distribution with parameter $n, (p_1, p_2, \ldots, p_k)$

Obs:

If H_0 is true then $\mathbb{E}(N_i)$=np_i^0

Test of hypothesis involving categorical data : Test Statistic

Test Statistic:

$$Q = \frac{\sum_{i=1}^{k}(N_i - np_i^0)^2}{np_i^0}$$

PEARSON'S CHI-SQUARED THEOREM

When H_0: $p_1 = p_{10}, \ldots, p_k = p_{k0}$ is true, the statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(N_i - np_{i0})^2}{np_{i0}}$$

has approximately a chi-squared distribution with $k - 1$ df. This approximation is reasonable provided that $np_{i0} \geq 5$ for every i ($i = 1, 2, \ldots, k$).

Test of hypothesis involving categorical data : Test Statistic

Test Statistic:

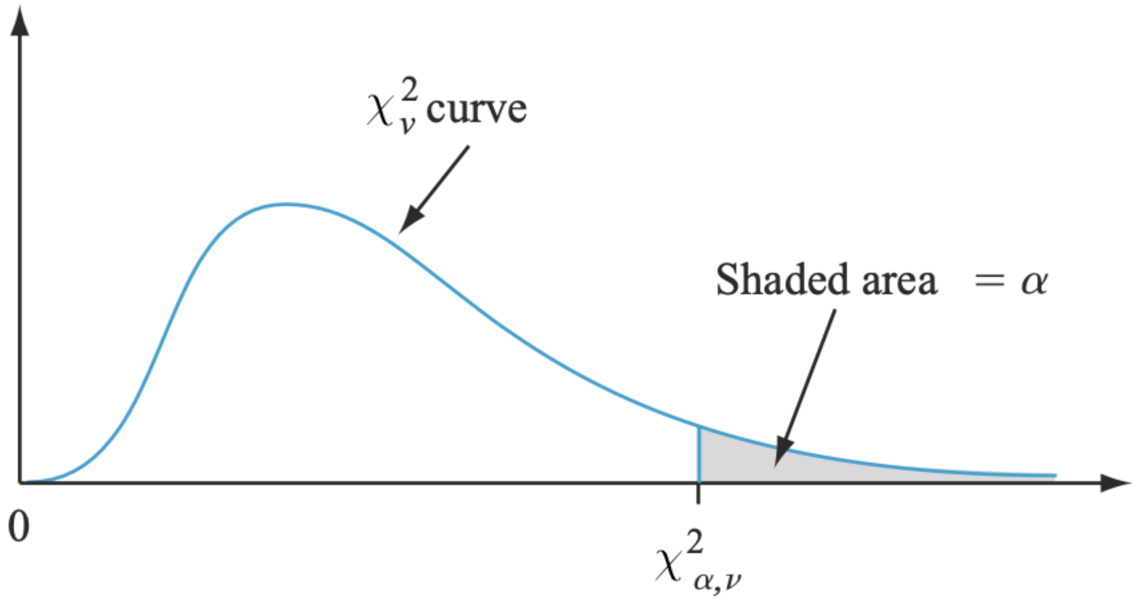
$$Q = \frac{\sum_{i=1}^{k}(N_i - np_i^0)^2}{np_i^0}$$

Few important points about Q:

1. Pearson Thm gives a very good approximation whenever $np_i^0 \geq 5, i = 1,2, \dots, k$.
2. However a satisfactory approximation is obtained when $np_i^0 \geq 1.5, i = 1,2, \dots, k$.

Q: Why is the denominator required in test?

Test of hypothesis involving categorical data : Test Procedure



Test of hypothesis involving categorical data : Test Procedure

CHI-SQUARED GOODNESS-OF-FIT TEST

Null hypothesis: H_0: $p_1 = p_{10}, \ldots, p_k = p_{k0}$

Alternative hypothesis: H_a: at least one p_i does not equal p_{i0}

Test statistic value:

$$\chi^2 = \sum_{i=1}^{k} \frac{(n_i - np_{i0})^2}{np_{i0}}$$

Rejection Region for Level α Test	P-value Calculation
$\chi^2 \geq \chi^2_{\alpha,k-1}$	area under χ^2_{k-1} curve to the right of the calculated χ^2

Test of hypothesis involving categorical data : Example

Table 10.1 Counts of blood types for white Californians

A	B	AB	O
2162	738	228	2876

Table 10.2 Theoretical probabilities of blood types for white Californians

A	B	AB	O
1/3	1/8	1/24	1/2

$$np_1^0 = 6004 \times \frac{1}{3} = 2001.3, \qquad np_2^0 = 6004 \times \frac{1}{8} = 750.5,$$

$$np_3^0 = 6004 \times \frac{1}{24} = 250.2, \quad \text{and} \quad np_4^0 = 6004 \times \frac{1}{2} = 3002.0.$$

The χ^2 test statistic is then

$$Q = \frac{(2162 - 2001.3)^2}{2001.3} + \frac{738 - 750.5}{750.5} + \frac{(228 - 250.2)^2}{250.2} + \frac{(2876 - 3002.0)^2}{3002.0} = 20.37.$$

Verify : p-val= 1.42×10^{-4}

$p - val = 1 - X_{k-1}^2$(Q) where X_{k-1}^2 is the c.d.f of chi-square distribution with k-1 d.f.

Test of hypothesis involving categorical data : Example

A TV station broadcasts a series of programs on the ill effects of smoking marijuana. After the series, the station wants to know whether people have changed their opinion about legalizing marijuana. Historical data from before the series showing the proportions of different categories of opinions is shown in the first table below, and the second table shows the sample proportion from the 500 randomly selected people.

Before the Series Was Shown

For legalization	Decriminalization	Existing law (fine or imprisonment)	No opinion
7%	18%	65%	10%

After the Series Was Shown

For legalization	Decriminalization	Existing law (fine or imprisonment)	No opinion
39%	9%	36%	16%

Here, $k = 4$, and we wish to test the following hypothesis

$$H_0: p_1 = 0.07; \quad p_2 = 0.18; \quad p_3 = 0.65; \quad p_4 = 0.1$$

Vs.

H_a: At least one of the probabilities is different from the hypothesized value.

Test of hypothesis involving categorical data : Example

Solution

We have the expected frequencies,

$$E_1 = (500)(0.07) = 35; \quad E_2 = 90; \quad E_3 = 325; \quad E_4 = 50.$$

The observed frequencies are

$$O_1 = (500)(0.39) = 195; \quad O_2 = 45; \quad O_3 = 180; \quad O_4 = 80.$$

The value of the test statistic is given by

$$Q^2 = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i}$$

$$= \left[\frac{(195-35)^2}{35} + \frac{(45-90)^2}{90} + \frac{(180-325)^2}{325} + \frac{(80-50)^2}{50}\right]$$

$$= 836.62.$$

From the χ^2-table, $\chi^2_{0.01,\,3} = 11.3449$. Because the test statistic $Q^2 = 836.62 > 11.3449$, we reject H_0 at $\alpha = 0.01$. Hence, the data suggest that people have changed their opinion after watching the series on the ill effects of smoking marijuana was shown. That is, the TV station broadcast did not change the opinion of the audience.

Test of hypothesis involving categorical data : Example

EXAMPLE 11.4.3

A die is rolled 60 times and the face values are recorded. The results of this experiment are:

Up face	1	2	3	4	5	6
Frequency	8	11	5	12	15	9

Is the die balanced fair? Test this question using $\alpha = 0.05$.

Solution
If the die is fair, we must have

$$p_1 = p_2 = \dots = p_6 = \frac{1}{6}$$

where p_i = P *(face value on the die is* i), i = 1, 2, ..., 6. *This experimental outcome follows the discrete uniform probability distribution.*

Test of hypothesis involving categorical data : Example

Hence,

$$H_0: p_1 = p_2 = \ldots = p_6 = \frac{1}{6}$$

Vs.

H_a: At least one of the probabilities is different from the hypothesized value of $1/6$

Note that $E^1 = n_1 p_1 = (60)\,(1/6) = 10, \ldots, E_6 = 10$, *and the condition of using this test is satisfied.*
We summarize the calculations in the following table:

Face value	1	2	3	4	5	6
Frequency, O_i	8	11	5	12	15	9
Expected value, E_i	10	10	10	10	10	10

The test statistic value is given by

$$Q^2 = \sum_{i=1}^{6} \frac{(O_i - E_i)^2}{E_i} = 6.$$

From the chi-square table with 5 d.f., $\chi^2_{0.05,\,5} = 11.070$.

Thus, $\chi^2_{0.05,5} = 11.070 = 11.07 > Q^2 = 6$ *and since the value of the test statistic does not fall in the rejection region, we do not reject* H_0. *Therefore, we do not have enough evidence to conclude that the die is not fair.*

Test of hypothesis: Testing Continuous Distribution

i. Partition the entire real line, or any particular interval that has probability 1, into a finite number k of disjoint subintervals. Generally, k is chosen so that the expected number of observations in each subinterval is at least 5 if H_0 is true.

ii. Determine the probability p_i^0 that the particular hypothesized distribution would assign to the ith subinterval, and calculate the expected number np_i^0 of observations in the ith subinterval ($i = 1, \ldots, k$).

iii. Count the number N_i of observations in the sample that fall within the ith subinterval ($i = 1, \ldots, k$).

iv. Calculate the value of Q as defined by Eq. (10.1.2). If the hypothesized distribution is correct, Q will have approximately the χ^2 distribution with $k - 1$ degrees of freedom.

$$Q = \frac{\sum_{i=1}^{k}(N_i - np_i^0)^2}{np_i^0}$$

Test of hypothesis: Testing Continuous Distribution

We wish to test the distribution that the log of lifetime of a ball-bearing follow. We hypothesize it to be normal with mean log(50)= 3.912 and variance 0.25. We have the following data which are the logarithm of lifetimes.

2.88	3.36	3.50	3.73	3.74	3.82	3.88	3.95
3.95	3.99	4.02	4.22	4.23	4.23	4.23	4.43
4.53	4.59	4.66	4.66	4.85	4.85	5.16	

Test of hypothesis: Testing Continuous Distribution

We wish to test the distribution that the log of lifetime of a ball-bearing follow. We hypothesize it to be normal with mean log(50)= 3.912 and variance 0.25. We have the following data which are the logarithm of lifetimes.

We find below the quantiles.

$$
\begin{aligned}
3.912 + 0.5\Phi^{-1}(0.25) &= 3.192 + 0.5 \times (-0.674) &&= 3.575, \\
3.912 + 0.5\Phi^{-1}(0.5) &= 3.192 + 0.5 \times 0 &&= 3.912, \\
3.912 + 0.5\Phi^{-1}(0.75) &= 3.192 + 0.5 \times 0.674 &&= 4.249,
\end{aligned}
$$

Value of the test statistic:

$$
\begin{aligned}
Q = \frac{(3 - 23 \times 0.25)^2}{23 \times 0.25} + \frac{(4 - 23 \times 0.25)^2}{23 \times 0.25} + \frac{(8 - 23 \times 0.25)^2}{23 \times 0.25} \\
+ \frac{(8 - 23 \times 0.25)^2}{23 \times 0.25} = 3.609.
\end{aligned}
$$

p-value = 0.307

Testing composite null hypothesis

H_0: There exists a value of $\theta \in \Omega$ such that
$p_i = \pi_i(\theta)$ for $i = 1, \ldots, k$,

H_1: The hypothesis H_0 is not true.

$$\theta = (\theta_1, \theta_2, \ldots, \theta_s);\ s < k - 1$$
$$\pi_i(\theta) > 0\ \forall i$$
$$\sum_1^k \pi_i(\theta) = 1\ \forall \theta \in \Omega$$

Testing composite null hypothesis: Example

Consider a gene that has two different alleles. Each individual in a given population must have one of three possible genotypes. If the alleles arrive independently from the two parents, and if every parent has the same probability θ of passing the first allele to each offspring, then then probabilities p_1, p_2 and p_3 of the three different genotypes can be represented in the following form:

$$H_0: \quad p_1 = \theta^2, \quad p_2 = 2\theta(1-\theta), \quad p_3 = (1-\theta)^2.$$

Here $\Omega = \{\theta \mid 0 < \theta < 1\}$

Verify that for $\theta \in \Omega$, $\pi_i(\theta)$ $\forall i$ satisfies the conditions given in the previous slide

Testing composite null hypothesis

H_0: There exists a value of $\theta \in \Omega$ such that
$p_i = \pi_i(\theta)$ for $i = 1, \ldots, k$,

H_1: The hypothesis H_0 is not true.

$$\theta = (\theta_1, \theta_2, \ldots, \theta_s);\ s < k - 1$$
$$\pi_i(\theta) > 0\ \forall i$$
$$\sum_1^k \pi_i(\theta) = 1\ \forall \theta \in \Omega$$

$$Q = \sum_{i=1}^{k} \frac{[N_i - n\pi_i(\hat{\theta})]^2}{n\pi_i(\hat{\theta})}.$$

$\hat{\theta}$ is the M.L.E. of θ

Thus we have to find the value of θ that maximizes the following likelihood based on the observation.

$$L(\theta) = \binom{n}{N_1, \ldots, N_k} [\pi_1(\theta)]^{N_1} \cdots [\pi_k(\theta)]^{N_k}.$$

$$\log L(\theta) = \log \binom{n}{N_1, \ldots, N_k} + \sum_{i=1}^{k} N_i \log \pi_i(\theta).$$

Testing composite null hypothesis: Example

Consider a gene that has two different alleles. Each individual in a given population must have one of three possible genotypes. If the alleles arrive independently from the two parents, and if every parent has the same probability θ of passing the first allele to each offspring, then then probabilities p_1, p_2 and p_3 of the three different genotypes can be represented in the following form:

$$\begin{aligned}\log L(\theta) &= N_1 \log(\theta^2) + N_2 \log[2\theta(1-\theta)] + N_3 \log[(1-\theta)^2] \\ &= (2N_1 + N_2) \log \theta + (2N_3 + N_2) \log(1-\theta) + N_2 \log 2\end{aligned}$$

$$\hat{\theta} = \frac{2N_1 + N_2}{2(N_1 + N_2 + N_3)} = \frac{2N_1 + N_2}{2n}.$$

Testing composite null hypothesis

H_0: There exists a value of $\theta \in \Omega$ such that
$p_i = \pi_i(\theta)$ for $i = 1, \ldots, k$,

H_1: The hypothesis H_0 is not true.

$$\theta = (\theta_1, \theta_2, \ldots, \theta_s);\ s < k - 1$$

$$\pi_i(\theta) > 0\ \forall i$$

$$\sum_{1}^{k} \pi_i(\theta) = 1\ \forall \theta \in \Omega$$

$$Q = \sum_{i=1}^{k} \frac{[N_i - n\pi_i(\hat{\theta})]^2}{n\pi_i(\hat{\theta})}.$$

$\hat{\theta}$ is the M.L.E. of θ

χ^2 Test for Composite Null. Suppose that the null hypothesis H_0 in (10.2.3) is true and certain regularity conditions are satisfied. Then as the sample size $n \to \infty$, the c.d.f. of Q in (10.2.4) converges to the c.d.f. of the χ^2 distribution with $k - 1 - s$ degrees of freedom. ■

Testing composite null hypothesis : Testing Independence

Contingency Tables. A table in which each observation is classified in two or more ways is called a *contingency table*.

Table 10.12 Classification of students by curriculum and candidate preference

	Candidate preferred			
Curriculum	A	B	Undecided	Totals
Engineering and science	24	23	12	59
Humanities and social sciences	24	14	10	48
Fine arts	17	8	13	38
Industrial and public administration	27	19	9	55
Totals	92	64	44	200

Testing composite null hypothesis : Testing Independence

Contingency Tables. A table in which each observation is classified in two or more ways is called a *contingency table*.

$$Q = \sum_{i=1}^{R} \sum_{j=1}^{C} \frac{(N_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}.$$

$$\hat{E}_{ij} = n\left(\frac{N_{i+}}{n}\right)\left(\frac{N_{+j}}{n}\right) = \frac{N_{i+}N_{+j}}{n}.$$

Row sum

Column Sum

$Q \sim \chi^2$ with $(R-1)(C-1)$ df

Testing composite null hypothesis : Testing Independence

Table 10.13 Expected cell counts for Example 10.3.2

Curriculum	Candidate preferred A	B	Undecided	Totals
Engineering and science	27.14	18.88	12.98	59
Humanities and social sciences	22.08	15.36	10.56	48
Fine arts	17.48	12.16	8.36	38
Industrial and public administrations	25.30	17.60	12.10	55
Totals	92	64	44	200

Verify: $Q = 6.68$

$p - value > 0.3$